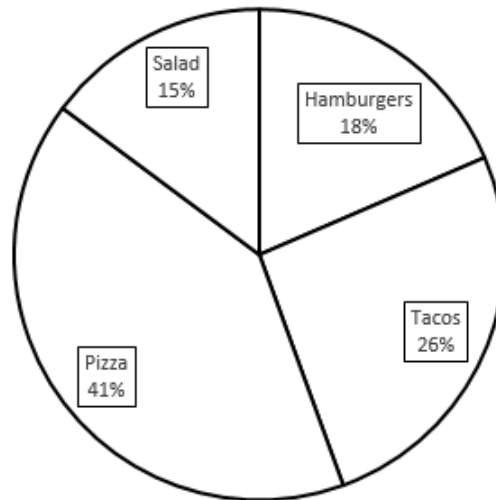


Algebra 1
Unit 12
Lesson 6 Practice

Name Answer Key
Date _____
Period _____

A circle graph is shown, with the results of a survey of middle school students about which lunch entrée they prefer, with a margin of error of 3%. There are 1250 students in the school.



1. Based on this survey, what is the estimated number of students who would prefer to have tacos?

$$0.26 \cdot 1250 = 325 \text{ students}$$

2. Based on this survey, what is the minimum number of students who would prefer to have hamburgers?

$$18\% - 3\% = 15\% \quad 0.15 \cdot 1250 \approx 187 \text{ students}$$

3. Based on this survey, what is the maximum number of students who would prefer to have salad?

$$15\% + 3\% = 18\% \quad 0.18 \cdot 1250 = 225 \text{ students}$$

4. Based on this survey, what is the range of students who would prefer to have tacos?

$$\begin{aligned} 26\% - 3\% &= 23\% & 0.23 \cdot 1250 &\approx 287 & \text{Approximately 287 to 363 students} \\ 26\% + 3\% &= 29\% & 0.29 \cdot 1250 &\approx 363 \end{aligned}$$

5. Based on this survey, what is the range of students who would prefer to have pizza?

$$\begin{aligned} 41\% - 3\% &= 38\% & 0.38 \cdot 1250 &= 475 & 475 \text{ to } 550 \text{ students} \\ 41\% + 3\% &= 44\% & 0.44 \cdot 1250 &= 550 \end{aligned}$$

A survey of parents indicated that 52% of students felt they have too much homework, with a margin of error of 2.1%.

6. Out of a group of 500 students, how many would likely feel that they have too much homework?

$$52 - 2.1 = 49.9 \quad 0.499 \cdot 500 \approx 249$$

$$52 + 2.1 = 54.1 \quad 0.541 \cdot 500 \approx 271$$

$$0.52 \cdot 500 = 260$$

$$52\% = 260 \text{ students}$$

$$\text{Range: } 249 \text{ to } 271 \text{ students}$$

7. Out of a group of 800 students, how many would likely not feel that they have too much homework?

$$100 - 52 = 48 \quad 0.48 \cdot 800 = 384$$

$$48 - 2.1 = 45.9 \quad 0.459 \cdot 800 \approx 367$$

$$48 + 2.1 = 50.1 \quad 0.501 \cdot 800 \approx 401$$

$$48\% = 384 \text{ students}$$

$$\text{Range: } 367 \text{ to } 401 \text{ students}$$

In a survey of 125 families, 24% reported that they eat out less than twice a week, 40% report they eat out twice a week, and 36% report they eat out more than twice a week. The survey has a margin of error of 1.5%.

8. What is the expected number of families that eat out less than twice a week?

$$0.24 \cdot 125 = 30$$

9. What is the range of families that eat out twice a week?

$$0.385 \cdot 125 \approx 48$$

$$48 \text{ to } 52$$

$$0.415 \cdot 125 \approx 52$$

10. What is the range of families that eat out less than twice a week?

$$0.225 \cdot 125 \approx 28$$

$$0.255 \cdot 125 \approx 32$$

$$28 \text{ to } 32$$